

ECG Denoising QRS Detection and Signal Quality Testbench

Technical Report | Educational Engineering Project

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Scope and conclusion. This report documents a reproducible Python testbench for ECG denoising, QRS timing, and signal-quality verification. It is not a diagnostic product and must not be used for patient care. In the verified 80-condition synthetic sweep, the aggressive configuration produced stronger QRS F1 than the balanced configuration, but it also removed more waveform bandwidth. The result supports only a synthetic timing-robustness claim, not clinical performance.

1 Engineering Contract

Item	Defined assumption
Input	One uniformly sampled ECG-like channel
Units	Millivolts and seconds
Sampling rate	250 Hz; Nyquist frequency 125 Hz
Verified duration	30 seconds with 35 exact reference peaks
Output	Filtered waveform, R peaks, RR and heart-rate summary, metrics and plots
Intended use	Education, prototyping, regression tests and portfolio evidence
Excluded use	Diagnosis, monitoring, alarms, treatment or medical decisions

2 Signal and Noise Model

The synthetic waveform is a sum of Gaussian P, Q, R, S and T components with small seeded beat-to-beat variability. Exact R-peak indices make one-to-one timing verification possible. The model is intentionally simple and is not a substitute for real physiology.

The artifact generator creates slow baseline drift, selectable 50 or 60 Hz interference, broadband random noise, low-pass random motion-like noise, and sparse impulsive electrode-like disturbances. Each condition is scaled to the requested total SNR with a fixed seed.

3 Processing Design

DC removal. Subtracting the record mean centers the trace but does not remove time-varying drift.

Notch filtering. A second-order IIR notch rejects narrow mains interference. Q equals center frequency divided by notch bandwidth.

Band-pass filtering. Butterworth second-order sections suppress baseline wander and broadband noise. The balanced path is 0.5 to 35 Hz, order 4; aggressive is 1 to 25 Hz, order 6.

Optional smoothing. The aggressive path applies a 21 ms Savitzky-Golay smoother. This improves timing robustness in the synthetic test but can attenuate morphology.

Phase choice. Forward-backward filtering cancels phase shift offline. It is noncausal, so real-time use requires causal filters and measured delay.

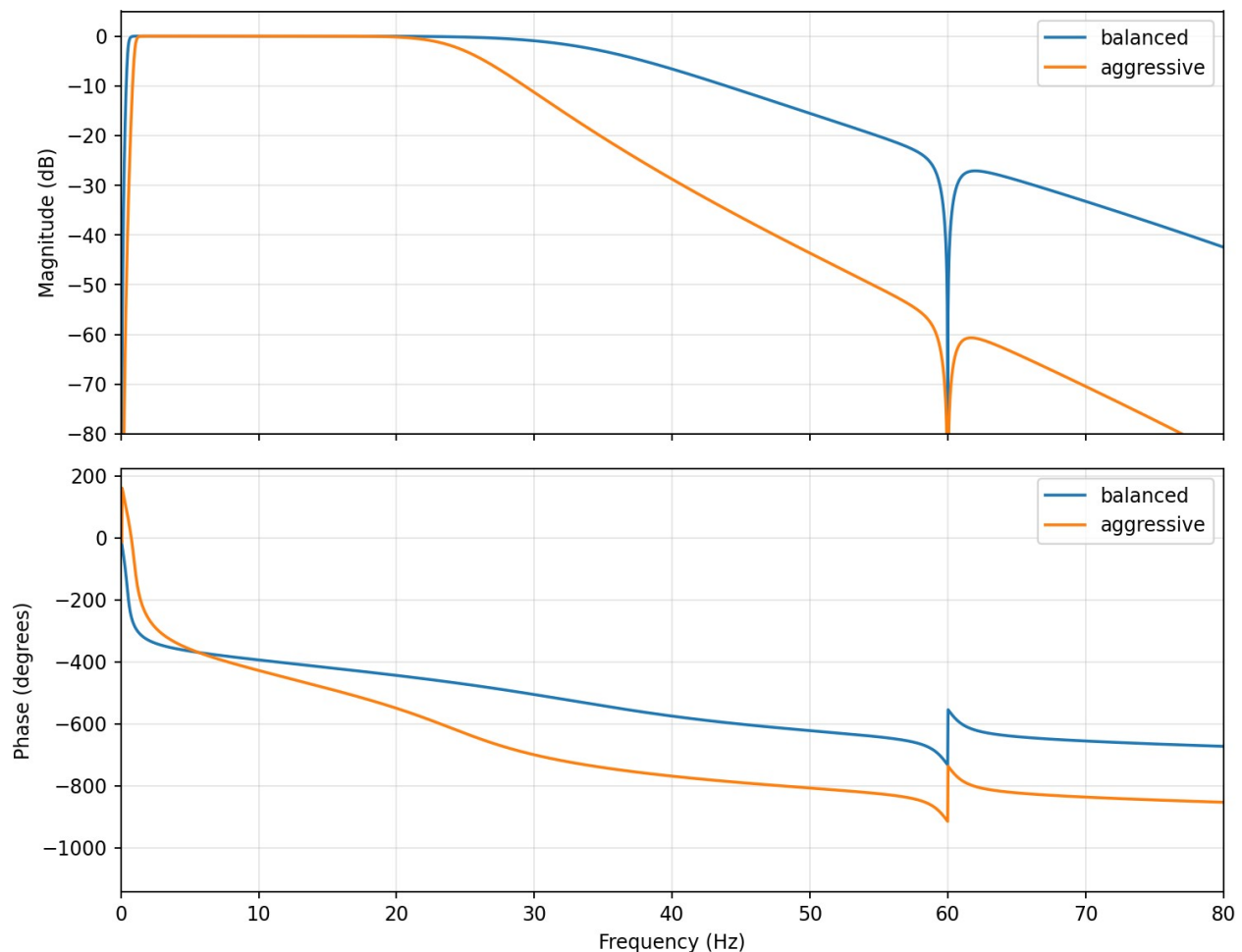


Figure 1. Combined notch and band-pass magnitude and phase responses.

4 QRS Detector

The explainable detector follows derivative, square, moving-window integration, robust threshold, refractory enforcement and local peak refinement. The 120 ms integration window collects QRS slope energy. A 250 ms refractory interval limits double detections. Each energy candidate is mapped back to the largest positive waveform sample within 120 ms.

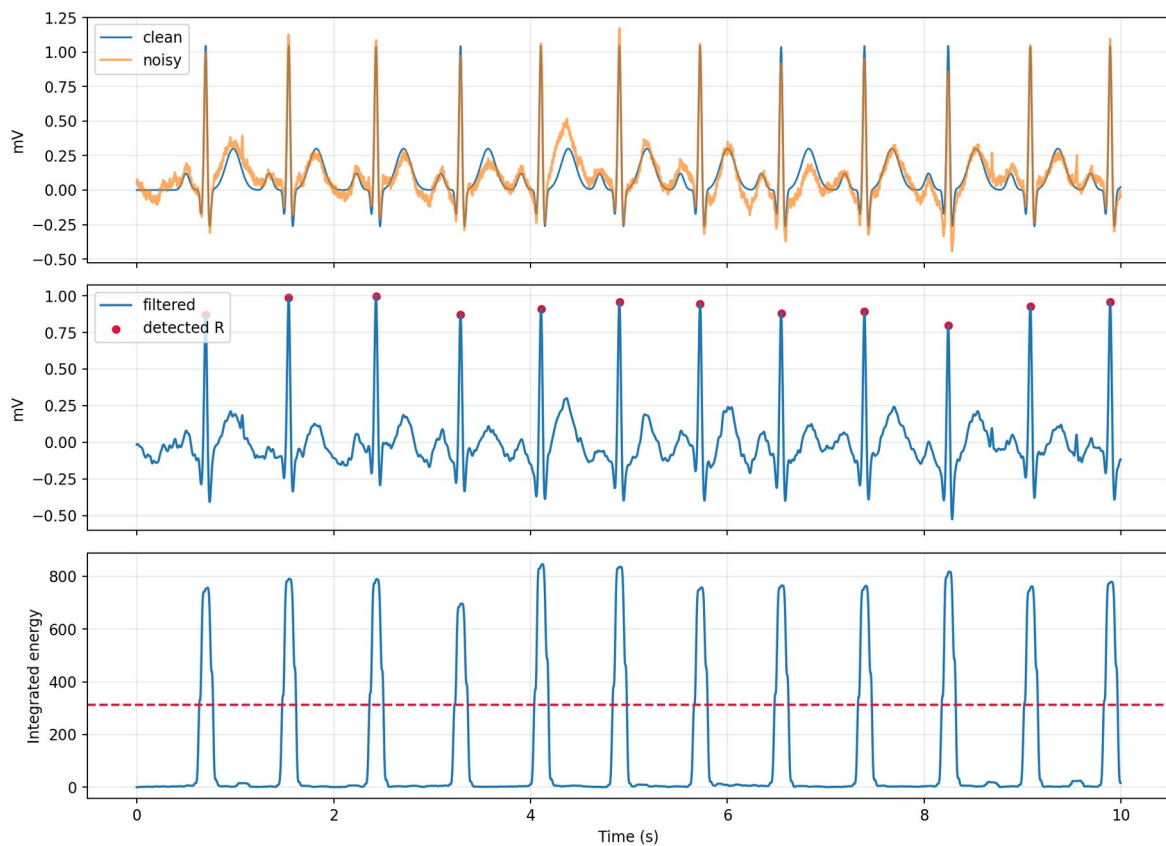


Figure 2. Clean and noisy input, filtered waveform with R detections, and integrated detector energy.

5 Metrics and Test Method

Where a clean reference exists, SNR compares reference power with residual error power and RMSE reports waveform error in millivolts. Each detection can match only one reference peak within plus or minus 100 ms. Unmatched detections are false positives and unmatched references are false negatives. Precision, recall and F1 summarize event detection. Timing error is evaluated only on matched pairs.

The signal-quality index is a transparent 0 to 1 heuristic combining plausible rate, RR consistency, waveform roughness and a clipping proxy. It is not clinically validated and has no medical pass threshold.

Sweep dimension	Values	Count
SNR	-5, 0, 5, 10, 20 dB	5
Artifact selection	all, baseline, motion, impulse	4
Mains frequency	50 and 60 Hz	2
Configuration	balanced and aggressive	2
Total	5 x 4 x 2 x 2	80

6 Verified Results

Metric	Balanced	Aggressive
Mean F1	0.9693	0.9898
Minimum F1	0.7576	0.8947
Mean precision	0.9664	0.9844
Mean recall	0.9729	0.9957
Mean absolute timing error ms	1.0192	0.8999
Mean output SNR dB	3.7393	4.1539
Mean heuristic SQI	0.7976	0.8311

The aggressive path had the higher mean and minimum F1 in this synthetic sweep. The sub-millisecond to approximately one-millisecond matched timing errors are expected to be optimistic because the synthetic R waves are sharp, zero-phase filtering is used, and candidates are refined to the waveform maximum.

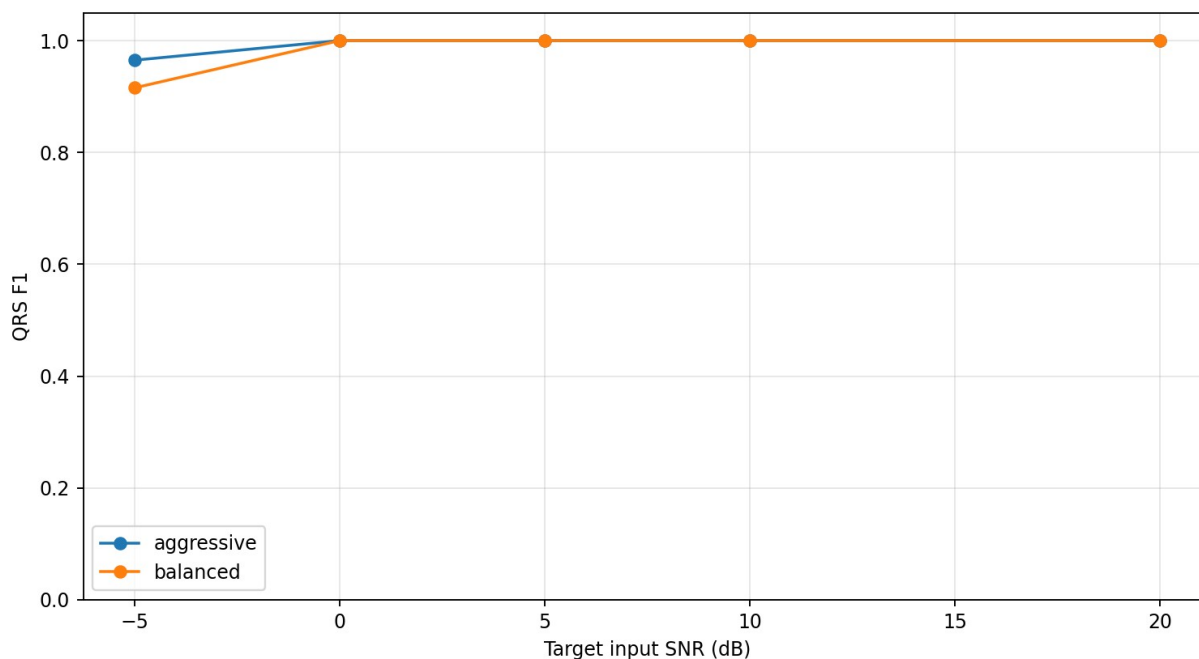


Figure 3. Mean QRS F1 versus requested SNR for the combined-artifact conditions.

7 Failure Analysis

Motion-like artifacts can produce sharp slopes that resemble QRS energy. At severe noise levels, they create false positives, hide true peaks or shift the local maximum. A global threshold is also weak when amplitude changes within a record. Edge padding can alter the first and last beats.

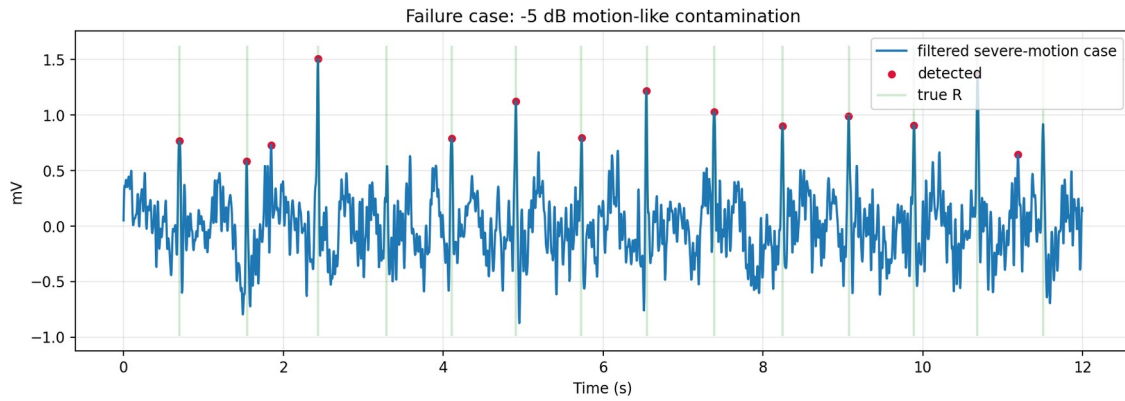


Figure 4. Severe motion-like condition showing detected events and exact reference timing.

8 Automated Verification

Test	Acceptance criterion
Filter stability	Notch and SOS poles lie inside the unit circle
Output integrity	Length preserved and all values finite
Reproducibility	Identical seed produces identical noise
Known signal	Precision and recall each at least 0.90
Input validation	Invalid Nyquist cutoff raises an error

All five tests passed in the delivered environment. The experiment runner also wrote a machine-readable manifest stating the seed, sampling rate, duration, beat count, row count and absence of public-record results.

9 Public Data Provenance

The optional extension targets the MIT-BIH Arrhythmia Database version 1.0.0 from PhysioNet. The source page describes 48 two-channel half-hour excerpts sampled at 360 Hz with reference beat annotations and lists the Open Data Commons Attribution License version 1.0. The verified package does not contain or claim public-record performance. Source:

<https://physionet.org/content/mitdb/1.0.0/>

10 Limitations and Next Work

The synthetic signal is regular and lacks the population, rhythm, lead, electrode and acquisition variability of real ECG. The detector uses a fixed global threshold. Offline zero-phase filtering hides causal delay. The SQI is heuristic. The next credible experiment is frozen-parameter, record-separated evaluation on annotated MIT-BIH records, followed by causal delay characterization, adaptive thresholds, explicit unusable-segment rejection and multi-lead agreement.

11 Interview Defense

Explain the project as a verification workflow: define a signal contract, inject named interference with fixed seeds, inspect filter response, use an explainable detector, match events one to one, compare two configurations, and show a failure case. State the key tradeoff clearly: the aggressive configuration improved synthetic timing robustness while risking greater morphology distortion. Never present these values as medical accuracy.